Problem Formulation

Define δ_i as:

If X ∈ ***class I*** and *misclassified as* ***Class II*** then *d(X)* will be **<0** ⇒ $\delta_i = -1$ ⇒ $d(X)\delta_i > 0$

If X ∈ ***class II*** and *misclassified as* ***Class I*** then *d(X)* will be **>0** ⇒ $\delta_i = +1$ ⇒ $d(X)\delta_i > 0$

Redefine Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors } (i)} \delta_i W^T X_i$$

⇒ *($W^T X_i$) takes positive and negative values and δ_i has the same sign*

*Summation is always **positive***

*Minimize the Error using **Gradient Descent** with step **η***

$$W^{(\text{updated})} = W^{(\text{old})} - \eta \frac{\partial \text{Error}}{\partial W} \quad \textbf{where} \quad \frac{\partial \text{Error}}{\partial W} = \sum_{\text{misclassified vectors}} \delta_i X_i$$

$$W^{(\text{updated})} = W^{(\text{old})} - \eta \sum_{\text{misclassified vectors ONLY } (i)} \delta_i X_i$$

Update after each sample

$$W^{(\text{updated})} = W^{(\text{old})} - \eta \delta_i X_i$$